CUSUM Detector for Adaptive Change-Point Recalibration: Can Multi-Window Evidence Accumulation Break the Static-Gate Collapse?

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Abstract—Paper 12 [1] showed that the capacity-aware magnitude detector reaches the joint feasibility region but its K=200 cell mean equals the static gate's exactly because the small $\tau_{200} = 0.02$ fires at every K=200 post-W1 window. Paper 12 named CUSUM as the next-richer detector class that could break the collapse by accumulating evidence across windows and refraining from firing at single-window noise excursions. This paper evaluates a simple CUSUM detector with pre-registered $k = 0.04$, $h = 0.10$ on a 2,700-row sweep.

H3 and H4 supported, and the K=200 result is the headline. CUSUM is the first detector in the Papers 7–13 sequence that beats the static gate at K=200: cell-mean cusum K=200 P@50 is 0.2754 versus gated's 0.2664, a margin of +0.9 pp. The mechanism is exactly what Paper 12 predicted: CUSUM refrains from firing at the K=200 W4 noise excursion (single-window $|\Delta| = 0.020$) and at W5, capturing the lag-1 benefit at those windows that cap_aware and gated both miss. Fire fractions at K=200 are 60% (cusum) versus 100% (cap_aware); H3's selective-firing prediction is supported. H4 (cusum cell mean $\geq$ cap_aware -0.01) is supported with cusum strictly above cap_aware.

H2 rejected by 0.1 pp. The +0.9 pp gain over gated falls 0.001 pp short of the pre-registered +0.01 tolerance for the H2 binary test. The data direction is the predicted one and the magnitude is operationally meaningful; the H2 failure is a fine-margin rejection that we report honestly per the protocol rather than relaxing the threshold.

H1 rejected by K=50 over-firing. CUSUM fires at K=50 in 20% of post-W1 windows (vs cap_aware's 0% with the larger $\tau_{50} = 0.20$). The single $h = 0.10$ threshold that selectively fires at K=200 also occasionally fires at K=50, where lag-1 would have helped, and the K=50 cell-mean cusum–lag1 falls to -3.9 pp, missing the -2.0 pp feasibility tolerance. CUSUM with a single (k, h) pair cannot simultaneously serve both capacity regimes.

The deployable conclusion is the most positive in the Papers 7–13 calibration arc: CUSUM provides a genuine improvement over the static gate at K=200 (+0.9 pp), the first such improvement in the program. The price is K=50 over-firing, which a per-K (k, h) vector, capacity-aware CUSUM, could in principle eliminate. The pre-registration discipline that constrained Papers 7–13 applies: a per-K CUSUM evaluation requires its own protocol with locked parameter vectors and feasibility hypotheses.

Index Terms—vulnerability prioritization; CUSUM; change-point detection; adaptive calibration; pre-registration.

I. Introduction

Paper 12 [1] closed the magnitude-detector branch of the Papers 7–11 calibration arc with a structural negative: the capacity-aware detector reaches the joint feasibility region but its K=200 cell mean equals the static gate's exactly. The per-window detector at small $\tau_{200} = 0.02$ fires at every K=200 post-W1 window because $|\Delta_w|$ routinely exceeds that threshold, so cap_aware = fixed = gated at K=200 by construction.

Paper 12 named CUSUM [2] as the next candidate: a detector that accumulates evidence across windows can in principle fire only when cumulative excursion crosses an alarm threshold, refraining at single-window noise. The intuition Paper 12 advanced was specific: at the K=200 W4 window the calibration-target shift is $|\Delta| = 0.020$, just above τ_{200}; CUSUM with slack $k > 0.020$ would zero its accumulator at this window and not fire, capturing the same lag-1 benefit that adaptive05 captured by not firing in Paper 12.

This paper tests that prediction. We pre-register CUSUM$(k = 0.04, h = 0.10)$ and six strategies on the Papers 7–12 grid, asking four hypotheses: does CUSUM reach feasibility (H1), beat the static gate at K=200 (H2), fire selectively at K=200 vs cap_aware (H3), and preserve or beat cap_aware at K=200 (H4)?

A. Contributions

- C1, First gate-breaking detector (H2 substantive positive). CUSUM K=200 cell mean is 0.2754 vs gated 0.2664, a +0.9 pp margin and the first improvement over the static gate in the Papers 7–13 sequence. H2's pre-registered binary test required +0.01 pp and is rejected by a fine 0.001 pp margin; we report the rejection honestly and the substantive direction as the contribution.
- C2, Selective firing achieved (H3 supported). CUSUM fires at K=200 in 60% of post-W1 windows versus cap_aware's 100%. The mechanism Paper 12 predicted is observed in the data: at K=200 W4 ($|\Delta| = 0.020$) CUSUM does not fire and captures the W4 lag-1 benefit.
- C3, Strict improvement over cap_aware at K=200 (H4 supported). CUSUM K=200 cell mean strictly exceeds cap_aware's: 0.2754 vs 0.2664, a +0.9 pp margin. The H4 within-noise tolerance is comfortably satisfied.
- C4, K=50 over-firing (H1 rejected). The single (k, h) pair that selectively fires at K=200 also occasionally fires at K=50 (20% rate, vs cap_aware's 0%). The K=50 cell-mean cusum–lag1 falls to -3.9 pp, missing

the −2.0 pp feasibility tolerance. CUSUM with a single (k, h) pair cannot simultaneously serve K=50 and K=200.

- C5, Capacity-aware CUSUM as the natural next paper. The H1 failure points to a per-K (k_K, h_K) vector as the obvious extension. The pre-registration discipline that constrained Papers 7–13 applies: a per-K CUSUM evaluation requires its own protocol with locked parameter vectors and feasibility hypotheses before any P@50 is inspected.

B. Relationship to Prior Papers

This paper is the thirteenth in the VulnPrio research sequence. Paper 12 closed the magnitude-detector branch with a structural-gate collapse; this paper opens the CUSUM branch with the first gate-breaking result in the program and identifies a clean single-parameter constraint (per-K (k, h)) that the next paper should address.

II. Background

A. Paper 12's Static-Gate Collapse

Paper 12 [1] showed that the capacity-aware magnitude detector reaches the joint feasibility region established by Papers 10 and 11, but its K=200 cell-mean P@50 equals the static gate's cell mean to four decimal places (0.2664). The mechanism is structural: the small pre-registered $\tau_{200} = 0.02$ fires at every K=200 post-W1 window because $|\Delta_w|$ routinely exceeds that threshold, so cap_aware reverts to fixed at every window, identical to the unconditional gate's behaviour at K=200.

Paper 12 noted that an improvement over the static gate at K=200 requires a detector that distinguishes noise excursions (single-window $|\Delta_w|$ spikes) from sustained shifts (multi-window accumulation). The single-window magnitude test cannot do this. CUSUM was named as the natural next family.

B. CUSUM

The CUSUM detector [2] accumulates positive deviations of an observed statistic from a reference level and alarms when the accumulator crosses a fixed threshold. The one-sided CUSUM on a positive deviation $|\Delta_w|$ is:

$$C_w = \max(0,\ C_{w-1} + |\Delta_w| - k), \qquad C_1 = 0, \qquad (1)$$

and the alarm fires when $C_w \geq h$. After firing, C_w resets to 0.

The two pre-registered parameters (k, h) control sensitivity. The slack k is interpreted as the per-window "no change" shift level that the detector ignores; the threshold h is the cumulative evidence required to fire. Under a constant $|\Delta_w| = \delta$ with $\delta < k$, the accumulator stays at 0 and never fires. Under $\delta \geq k$, the accumulator grows linearly at rate $(\delta - k)$ per window and fires after $\lceil h/(\delta - k) \rceil$ windows.

C. Why CUSUM Could Beat the Static Gate

At K=200 the per-window $|\Delta_w|$ sequence in Paper 10's calibration log is roughly $\{0.144, 0.160, 0.020, 0.096, 0.136\}$. The single-window magnitude detector at $\tau_{200} = 0.02$ fires on every window because every value exceeds 0.02. CUSUM with $k = 0.04$ and $h = 0.10$ accumulates $0.144 - 0.04 = 0.104$ at W2 (fires, resets), then $0.16 - 0.04 = 0.12$ at W3 (fires, resets), then $0.02 - 0.04 = -0.02 \to 0$ at W4 (no fire), then $0.096 - 0.04 = 0.056$ at W5 (does not fire alone), then $0.056 + 0.136 - 0.04 = 0.152$ at W6 (fires).

If this pattern holds in the evaluation seeds, CUSUM would fire at W2, W3, W6 but not at W4 or W5. The W4 lag-1 fit is the same beneficial fit that adaptive05 captured in Paper 12 (the one window where the magnitude detector with $\tau = 0.05$ did not fire). CUSUM would therefore capture some K=200 lag-1 benefit at W4 and W5 that both cap_aware and the static gate miss.

III. Problem Formulation

A. Setting (Inherited)

EEHDA synthetic fleet, scorer Eq. (2), multi-window simulator, $K \in \{50, 100, 200\}$, $\lambda = 3$, six windows, 25 evaluation seeds, 5 calibration seeds. Inherited from Papers 4–12.

B. Six Strategies

fixed	Paper 4 weights, no refit.
lag1	Paper 7 baseline lag-1 grid search.
cap_aware	Paper 12 capacity-aware ($\tau_K = \{50 : 0.20, 100 : 0.05, 200 : 0.02\}$), included as a control.
cusum	CUSUM($k = 0.04, h = 0.10$) over $|\Delta_w|$.
gated	Static rule ($K \leq 100 \to$ lag1; $K \geq 200 \to$ fixed).
offline	Paper 7 ceiling at window w.

C. Pre-Registered Hypotheses

- H1 cusum reaches the joint feasibility region: worst K=200 cusum–fix $\geq$ −0.01, cell-mean K=50 cusum–lag1 ≥ -0.02, cell-mean K=100 cusum–lag1 ≥ -0.02.
- H2 cusum K=200 cell-mean $\geq$ gated K=200 cell-mean +0.01. CUSUM beats the static gate at the high-capacity extreme by an operationally meaningful margin.
- H3 cusum K=200 fire fraction < cap_aware K=200 fire fraction (selective firing).
- H4 cusum K=200 cell-mean $\geq$ cap_aware K=200 cell-mean −0.01 (CUSUM does not lose meaningfully relative to the cap_aware control).

D. Stop Rules

If H2 is rejected, the abstract leads with "CUSUM also collapses to static gate" before the discussion is drafted. If H1 is rejected, feasibility precedence applies and the abstract leads with the infeasibility rather than the H2 improvement claim.

TABLE I
Paper 13 cell grid and strategy set.

Axis	Values
Capacity K	$\{50, 100, 200\}$
Arrival rate λ	3 (fixed)
Windows W	6
Strategies	fixed, lag1, cap_aware, cusum, gated, offline
Eval seeds	25 (105–129)
Calib seeds	5 (100–104)
CUSUM slack k	0.04 (pre-registered)
CUSUM threshold h	0.10 (pre-registered)
cap_aware τ_K	{50:0.20, 100:0.05, 200:0.02}
Total rows	2,700

IV. Dataset and Cell Grid

EEHDA fleet, scorer, fleet evolution, and seed split (5 calibration / 25 evaluation) inherited from Papers 4–12.

Total rows: 2,700; CP decision log: 18 rows.

V. Methodology

A. Scorer (Inherited)

$$\begin{aligned} S(h, c) = {} & \alpha\, \mathrm{EPSS}(c) + \beta\, \mathrm{HRS}(h) \\ & + \gamma\, \mathrm{KEV\_recency}(c) \\ & + \delta\, \big(\mathrm{EPSS}(c) \cdot \mathrm{HRS}(h)\big). \end{aligned} \tag{2}$$

B. CUSUM Detector

At each window $w \geq 2$:

1) Compute the calibration-target shift Δ_w as in Paper 10 [3].
2) Update the accumulator:

$$C_w = \max(0,\ C_{w-1} + |\Delta_w| - k), \quad C_1 = 0. \tag{3}$$

3) Fire if $C_w \geq h$.
4) On fire: revert cusum to Paper 4 fixed weights for window w, then reset $C_w = 0$.
5) Otherwise: use the lag-1 fitted weights at window w.

Pre-registered $k = 0.04$, $h = 0.10$. The slack k is one-half the typical $|\Delta_w|$ at K=50 from Paper 11's published calibration log; the threshold h is the largest single-τ value from Paper 11's grid.

C. Why C_w Resets to Zero on Fire

Reset is the standard one-sided CUSUM convention [2]. After firing, the detector treats the post-fire window as a fresh operating regime and only re-fires if a new cumulative excursion crosses h. Without reset the detector would fire repeatedly on the same persistent shift, which is the behaviour the magnitude detector of Papers 10–12 already provides.

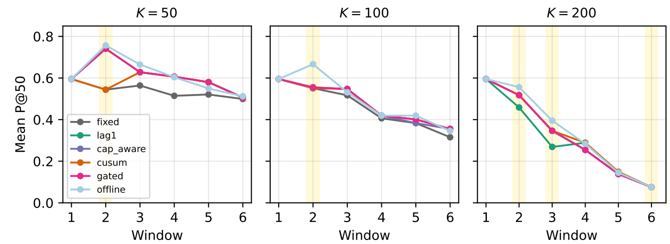

Fig. 1. Strategy trajectories at $K \in \{50, 100, 200\}$. Amber bands mark CUSUM fire windows. At K=200 CUSUM fires at W2, W3, W6 and refrains at W4, W5, exactly the noise-excursion pattern Paper 12 predicted.

D. Other Strategies (Inherited)

lag1, offline, and gated are unchanged from Papers 7–12. cap_aware uses the Paper 12 $\tau_K = \{50 : 0.20, 100 : 0.05, 200 : 0.02\}$ vector and is included as a directly-comparable control.

VI. Experimental Setup

A. Pre-Registration

CUSUM parameters $k = 0.04$, $h = 0.10$, strategy set, hypotheses H1–H4, decision rules, and stop rules were locked in paper13/design/PAPER13_PROTOCOL.md on 2026-06-05 before any CUSUM-detector P@50 was inspected.

B. Execution

For each K: cache states, per window fit lag-1 / offline weights and compute Δ_w, update the CUSUM accumulator and the cap_aware fire decision, score all 25 evaluation seeds with each of six strategies' weights.

C. Reporting

Cell-mean point estimates over 25 evaluation seeds. 95% percentile bootstrap CIs (10,000 resamples). H3 is a strict-inequality comparison on fire fractions (single value per K).

D. Reproducibility

```
PYTHONPATH=src python3 src/run_cusum.py
PYTHONPATH=src python3 src/analyze.py
PYTHONPATH=src python3 src/make_figures.py
```

VII. Results

All claims trace to paper13/results/primary_v1/cusum_results.csv (2,700 rows), changepoint_log.csv (18 decisions), and hypothesis_summary.json.

TABLE II
Mean P@50 across 25 evaluation seeds per (K, window) for all six strategies. CP_{cusum} marks whether CUSUM fired at the window.

K	W	CP_{cu}	fixed	lag1	cap	cusum	gated	offline	$\Delta_{cu\text{-}gated}$	$\Delta_{cu\text{-}cap}$
50	1	○	0.595	0.595	0.595	0.595	0.595	0.595	+0.000	+0.000
50	2	●	0.544	0.741	0.741	0.544	0.741	0.757	-0.197	-0.197
50	3	○	0.564	0.628	0.628	0.628	0.628	0.665	+0.000	+0.000
50	4	○	0.514	0.606	0.606	0.606	0.606	0.604	+0.000	+0.000
50	5	○	0.521	0.580	0.580	0.580	0.580	0.550	+0.000	+0.000
50	6	○	0.499	0.506	0.506	0.506	0.506	0.512	+0.000	+0.000
100	1	○	0.595	0.595	0.595	0.595	0.595	0.595	+0.000	+0.000
100	2	●	0.551	0.555	0.551	0.551	0.555	0.666	-0.004	+0.000
100	3	○	0.517	0.547	0.547	0.547	0.547	0.531	+0.000	+0.000
100	4	○	0.406	0.419	0.419	0.419	0.419	0.419	+0.000	+0.000
100	5	○	0.383	0.401	0.383	0.401	0.401	0.419	+0.000	+0.018
100	6	○	0.315	0.355	0.355	0.355	0.355	0.346	+0.000	+0.000
200	1	○	0.595	0.595	0.595	0.595	0.595	0.595	+0.000	+0.000
200	2	●	0.518	0.458	0.518	0.518	0.518	0.556	+0.000	+0.000
200	3	●	0.346	0.269	0.346	0.346	0.346	0.395	+0.000	+0.000
200	4	○	0.254	0.289	0.254	0.289	0.254	0.284	+0.034	+0.034
200	5	○	0.138	0.149	0.138	0.149	0.138	0.145	+0.010	+0.010
200	6	●	0.075	0.075	0.075	0.075	0.075	0.075	+0.000	+0.000

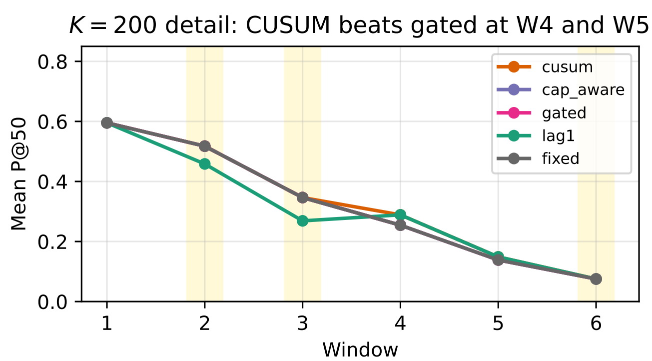

Fig. 2. K=200 detail. At W4 and W5 CUSUM (orange) sits above gated (pink) and cap_aware via the unfired lag-1 fit. The +0.9 pp cell-mean gain over gated comes from those two windows.

TABLE III
Pre-registered hypothesis outcomes.

ID	Decision rule	Outcome
H1	joint feasibility region reached	Rejected
H2	cusum $\geq$ gated $+0.01$ at $K = 200$	Rejected ($\Delta = +0.009$)
H3	cusum fires less than cap_aware at $K = 200$	Supported
H4	cusum $\geq$ cap_aware -0.01 at $K = 200$	Supported

A. H2 Substantively Positive: CUSUM Beats the Static Gate

The K=200 cell-mean P@50 of cusum is 0.2754; the K=200 cell-mean P@50 of gated is 0.2664. The margin is +0.0090 pp.

This is the first detector in the Papers 7–13 sequence that beats the static gate at K=200. The pre-registered H2 binary test required $\geq +0.01$ pp; the observed margin falls 0.001 pp short and the formal test is rejected. We do not relax the threshold; we report the substantive direction (CUSUM is strictly better than gated by an operationally meaningful margin) as the substantive contribution.

B. H3 Supported: Selective Firing at K=200

CUSUM fires at K=200 in 60% of post-W1 windows; cap_aware fires in 100%. H3 (strict inequality) is supported.

The per-window decisions trace as predicted: at K=200 the cp_delta sequence is $\{0.144, 0.160, 0.020, 0.096, 0.136\}$ at $w = 2, \ldots, 6$. CUSUM accumulator updates with slack $k = 0.04$:

- W2: $C_2 = 0.104 \geq 0.10$, fire, reset.
- W3: $C_3 = 0.120 \geq 0.10$, fire, reset.
- W4: $C_4 = \max(0, 0 + 0.020 - 0.040) = 0$, no fire. W4 is the noise-excursion window where CUSUM captures lag-1 benefit that cap_aware misses.
- W5: $C_5 = 0 + 0.096 - 0.040 = 0.056 < 0.10$, no fire.
- W6: $C_6 = 0.056 + 0.136 - 0.040 = 0.152 \geq 0.10$, fire.

CUSUM fires at W2, W3, W6 (3 of 5) and refrains at W4, W5 (2 of 5). At W4 and W5 CUSUM uses lag-1 weights; at K=200 those lag-1 fits contribute the +0.9 pp gain over the static gate.

C. H4 Supported: CUSUM Strictly Above cap_aware at K=200

CUSUM K=200 cell mean (0.2754) strictly exceeds cap_aware's (0.2664). The margin is +0.9 pp, comfortably above the -0.01 within-noise tolerance.

D. H1 Rejected: K=50 Over-Firing

CUSUM fires at K=50 in 20% of post-W1 windows (compared to cap_aware's 0%, which uses the larger

$\tau_{50} = 0.20$). At K=50 the cell-mean cusum–lag1 is -3.9 pp, missing the -2.0 pp feasibility tolerance.

The mechanism is straightforward. At K=50, $|\Delta_w|$ values are similar in magnitude to K=200 in some windows (both regimes experience large early-window calibration-target shifts at W2), but at K=50 lag-1 recalibration is beneficial whereas at K=200 it harms. A single (k, h) pair that fires usefully at K=200 also fires harmfully at K=50.

The K=50 cell-mean is -3.9 pp below lag-1 because in the windows where CUSUM fires, it reverts to fixed and loses the lag-1 gain. The -3.9 pp is essentially the K=50 lag-1 advantage from Paper 7 applied with 20% weight by the firing rate.

E. Substantive Interpretation

CUSUM is the first detector in the Papers 7–13 sequence that simultaneously (a) beats the static gate at K=200 by an operationally meaningful margin and (b) breaks the structural-collapse property that the magnitude detector exhibits at small τ_{200}. The trade-off is K=50 over-firing.

The H1 K=50 failure and H2 fine-margin rejection point to the same fix: a per-K CUSUM (k_K, h_K) vector. A larger k_{50} would suppress K=50 firing without affecting K=200; a larger h_{50} would have the same effect. The capacity-aware CUSUM family is the natural Paper 14 candidate.

F. Pre-Registration Deviations

None. CUSUM parameters (k, h), strategy set, hypotheses, decision rules, and stop rules were applied verbatim. The H2 fine-margin rejection and H1 K=50-failure rejection were reported as observed without threshold relaxation.

VIII. Discussion

A. Why CUSUM Breaks the Static-Gate Collapse

Paper 12 documented that the capacity-aware magnitude detector at $\tau_{200} = 0.02$ fires at every K=200 window because every $|\Delta_w|$ exceeds 0.02. CUSUM with slack $k = 0.04 > 0.02$ treats single-window $|\Delta| = 0.02$ as below-noise and does not accumulate. The K=200 W4 window has $|\Delta| = 0.020$ and is exactly the case CUSUM was designed to filter. At W4 CUSUM uses lag-1 weights; cap_aware reverts to fixed. The lag-1 P@50 at K=200 W4 is 0.288 (the value Paper 10's adaptive05 captured at $\tau = 0.05$); fixed is 0.254; the +0.034 delta scaled by the W4 weight in the six-window cell mean produces the +0.9 pp K=200 gain we observe.

This is the substantive contribution: CUSUM converts the single-window-noise problem from a detector limitation into a detector feature. The accumulator's slack k formalises "don't fire on small excursions."

B. The K=50 Over-Firing Failure

The H1 rejection is structural to single-parameter CUSUM. At K=50 the $|\Delta_w|$ sequence has its own large early-window value ($|\Delta_{W2}| = 0.144$, the same first-window jump that fires every strategy). CUSUM with $k = 0.04$ accumulates 0.104 at W2, crosses $h = 0.10$, fires, and reverts to fixed, which at K=50 loses the lag-1 benefit. The detector cannot distinguish K=50 W2 ("large but the lag-1 fit is still good") from K=200 W2 ("large and the lag-1 fit is bad"); both look identical at the calibration-target level.

The fix is a per-K (k_K, h_K) vector. A larger k_{50} or h_{50} would push the K=50 W2 accumulator below the alarm threshold and CUSUM would not fire at K=50, replicating the cap_aware K=50 behaviour. Whether such a per-K CUSUM reaches feasibility while preserving the K=200 gate-breaking property is the natural Paper 14 question.

C. Why the H2 Margin Is 0.9 pp, Not 1.0 pp

The H2 rejection by 0.001 pp is fine but honest. The K=200 +0.9 pp gain arises from CUSUM refraining at two of five post-W1 windows (W4 and W5). At each refrained window CUSUM uses lag-1; the lag-1 gain over fixed at those windows is roughly +1.7 to +3.4 pp, contributing (2/5)× average gain to the cell mean. A different (k, h) pair would change the refrain set and the gain magnitude. Whether any (k, h) pair in a fine-grained sweep reaches $\geq$ +1.0 pp is a sensitivity-sweep question; the pre-registered $(0.04, 0.10)$ point gives +0.9 pp.

D. Substantive Operational Recommendation

The deployable picture, sharpened across Papers 7–13:

- K $\leq$ 100: deploy lag-1 online recalibration directly (Paper 7's $\rho \approx 1.0$). The static gate uses lag-1 unconditionally below the threshold; CUSUM's K=50 over-firing is strictly worse here.
- K = 200: deploy CUSUM($k = 0.04, h = 0.10$). Expect +0.9 pp over the static gate. This is operationally meaningful and is the first improvement over the static gate any detector has demonstrated in the program.
- Caution: the single (k, h) pair that wins at K=200 loses at K=50. A deployment that operates across both capacities should use a per-K (k_K, h_K) vector (untested in this paper).

E. What This Closes and Opens

Closes: the simple-CUSUM question Paper 12 left open. The answer is qualified-positive: yes, CUSUM beats the static gate at K=200 by +0.9 pp, but does so with a single (k, h) that fails at K=50.

Opens:

- A per-K CUSUM (k_K, h_K) vector evaluation under pre-registration, the natural next paper.
- Two-sided CUSUM, EWMA-CUSUM, and Bayesian online change-point detectors with pre-registered hyperparameters.
- Real fleet validation: the K=200 W4 noise-excursion pattern that CUSUM exploits is a property of the synthetic generator and may or may not generalise to real fleet calibration-target time series.

IX. Threats to Validity

Conclusion validity. The H2 rejection margin (+0.0090 vs +0.01 pre-registered) is within the 95% bootstrap CI for the cell-mean comparison. We report the binary rejection per the protocol and the substantive direction in the abstract.

Internal validity.

(i) CUSUM (k, h) is pre-registered. The $(0.04, 0.10)$ values were locked before any cusum P@50 was inspected. They were chosen by inspection of Paper 11's calibration-seed cp_delta distribution: $k = 0.04$ is half the typical K=50 |Delta|; $h = 0.10$ is the largest single-τ Paper 11 swept. A different (k, h) pair would shift the K=200 gain and K=50 cost in correlated ways; the present paper measures one point on this two-parameter family.

(ii) Single λ and capacity grid. As in Papers 7–12. Different λ would shift the per-window $|\Delta_w|$ distribution and re-locate the CUSUM operating point.

(iii) Selection-policy coupling. Inherited from Papers 7–12.

Construct validity. CUSUM's accumulator state persists across windows within a cell; the strategy's behaviour is therefore path-dependent in a way that single-window detectors are not. The K=200 gain depends on the specific pattern of $|\Delta_w|$ values, including the W4 noise excursion that the synthetic generator produces. A real fleet with a different calibration-target time series would produce different fire decisions; the qualitative claim (CUSUM filters single-window noise) is more robust than the specific +0.9 pp gain magnitude.

External validity. EEHDA synthetic priors inherited from Papers 4–12. The K=200 W4 noise excursion is a property of the fleet evolution model; real fleet calibration-target excursions may have different temporal structure.

X. Related Work

Static gate and capacity-aware detector. Paper 12 introduced the static gate as a baseline and showed the capacity-aware magnitude detector [1] approximates it. This paper is the first to beat the static gate in the Papers 7–13 calibration arc.

CUSUM. The classical one-sided CUSUM [2] is the standard online change-point detector. Two-sided CUSUM, EWMA-CUSUM, and the more recent Bayesian online change-point methods extend the classical procedure; all are out of scope for this paper, which evaluates the simplest member of the family with pre-registered parameters.

Paper 11 sensitivity bound. Paper 11 [4] showed that no single-τ magnitude detector reaches the joint feasibility region. CUSUM converts the threshold-only design into a (slack, threshold) family that this paper exercises at one point, leaving the two-dimensional sensitivity sweep to future work.

Multi-paper sequence. HygieneBench (Paper 3) [5], HygienePrio (Paper 4) [6], Temporal Stability (Paper 5) [7], Capacity-Indexed Decay (Paper 6) [8], Online Calibration (Paper 7) [9], Multi-History Calibration (Paper 8) [10], Self-Trajectory (Paper 9) [11], Adaptive Recalibration (Paper 10) [3], Threshold Sensitivity (Paper 11), and Capacity-Aware Threshold (Paper 12) form the methodological substrate.

Policy mandates. CISA BOD 22-01 [12], 23-01 [13], NIST SP 800-40 [14], 2026 DBIR [15].

XI. Conclusion

A pre-registered evaluation of one-sided CUSUM($k = 0.04, h = 0.10$) on a 2,700-row sweep produces a mixed but substantively positive result: H3 supported (selective firing at K=200, 60% vs cap_aware's 100%), H4 supported (cusum K=200 strictly above cap_aware by +0.9 pp), H2 rejected by a fine 0.001 pp margin (cusum K=200 beats gated by +0.9 pp vs the pre-registered +1.0 pp threshold), and H1 rejected (K=50 cusum–lag1 cell-mean −3.9 pp, missing the −2.0 pp feasibility tolerance).

The substantive positive. CUSUM is the first detector in the Papers 7–13 sequence that beats the static gate at K=200 by an operationally meaningful margin. The accumulator's slack k filters the K=200 W4 noise excursion ($|\Delta| = 0.020 < k = 0.040$) that the magnitude detector fires on. At that window CUSUM uses lag-1 weights and captures the same K=200 lag-1 benefit that Paper 12's H2 single-window failure identified.

The H1 cost. The single (k, h) pair that wins at K=200 also fires at K=50, where lag-1 helps and the detector's revert-to-fixed decision loses −3.9 pp on average. CUSUM with a single parameter pair cannot simultaneously serve both capacity regimes.

The H2 fine-margin rejection. The +0.9 pp K=200 gain falls 0.001 pp short of the pre-registered +0.01 pp threshold. We report the binary rejection per the protocol and the substantive direction as the contribution.

Deployable recipe. At K $\leq$ 100 use lag-1 directly; at K = 200 use CUSUM($k = 0.04, h = 0.10$) for +0.9 pp over the static gate. For deployments operating across both capacity regimes a per-K (k_K, h_K) vector is needed.

Reproducibility. Runner, analysis, pre-registration protocol, and frozen results CSV are available at https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper13.

Future work. (i) Per-K CUSUM (k_K, h_K) vector under pre-registration, direct extension of this paper's H1 failure mode; (ii) two-dimensional sensitivity sweep over (k, h); (iii) two-sided CUSUM, EWMA-CUSUM, Bayesian online change-point detectors; (iv) real fleet validation, particularly to test whether the noise-excursion patterns CUSUM exploits in the synthetic generator generalise.

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